Harpreet Singh

PhD Research Assistant | Computational Behavioural Neuroscience | Embedded AI / FPGA

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Profile

PhD researcher in computational behavioural neuroscience at the University of Lethbridge, working as a Research Assistant with hands-on experience in embedded systems, FPGA-based AI acceleration, mixed-signal acquisition, and Raspberry Pi/Linux integration. Built Vitis HLS and Vivado starter systems for temporal SNN, QiSNN, and ANN neural-network accelerators; implemented MicroBlaze, BRAM, UDP, and UART observability flows; and developed Raspberry Pi software for voice-controlled Linux operation and EEG/ADC/DAC hardware testing.

Technical Skills

FPGA and hardware	Xilinx Vivado, Vitis HLS, MicroBlaze, AXI4-Lite, BRAM maps, UARTLite, Ethernet/UDP, Arty A7-35T, Arty A7-100T, Arty Z7, Artix-7, fixed-point design
Embedded Linux	Raspberry Pi 5, Python, C/C++, SPI, GPIO, threading, shell tooling, systemd setup, Linux source exploration, command-line diagnostics
AI and acceleration	MNIST, N-MNIST, temporal SNNs, QiSNN, ANN baselines, Keras, PyTorch, quantized weights, generated test benches, model-to-HLS export
Data acquisition	ADS1299 8-channel EEG, ADS8688 ADC, DAC8565, MUX/TDM sequencing, CSV logging, real-time matplotlib plots, hardware validation scripts

Current Research Role

Research Assistant and PhD Student | *University of Lethbridge* | *Computational Behavioural Neuroscience* | *Present*

- Conducting doctoral research in computational behavioural neuroscience with emphasis on neural data, behaviour-linked computation, and hardware-aware analysis workflows.
- Building research software and hardware-support tools across FPGA neural acceleration, embedded Linux, Raspberry Pi data acquisition, and biomedical signal processing.

Selected Engineering Work

fpgaBrain Multi-FPGA Neural-Network Observatory | *N-MNIST / FPGA AI / UDP telemetry*

- Designed a multi-board experiment flow where the same N-MNIST event sample is streamed to several FPGA neural networks and compared through host-side fusion.
- Defined a temporal input contract of 12 time bins x 196 features, packaged as 2352 signed fixed-point words with 10 output class scores.
- Created host utilities for board mapping, UDP streaming, dry-run validation, and YES/NO/UNCERTAIN decision fusion from UART observation logs.

Active QiSNN Temporal N-MNIST FPGA Pipeline | *Vitis HLS / Vivado / MicroBlaze*

- Separated a true temporal QiSNN path from a collapsed-input path so each HLS step consumes input[t * 196 + j], matching the PyTorch training loop.
- Mapped 2352 temporal input words and 10 output scores into BRAM windows with AXI4-Lite ap_ctrl_hs control and ap_fixed<16,6> payload handling.

A100T 1534N Binary LIF Observer | *Artix-7 / LIF accelerator / decision telemetry*

- Built a starter design for the Arty A7-100T around a 1534-neuron binary LIF network with a 1024 -> 256 -> 128 -> 64 -> 32 -> 16 -> 8 -> 4 -> 2 topology.
- Defined input, output, and state BRAM maps plus a MicroBlaze UDP/UART observer that reports prediction, latency, output scores, and hidden-layer spike/margin probes.

Raspberry Pi EEG_BCT Mixed-Signal Platform | *ADS1299 / ADS8688 / DAC8565 / SPI-GPIO control*

- Developed Raspberry Pi Python scripts for an 8-channel ADS1299 EEG acquisition path, 8-channel ADS8688 ADC monitoring, and DAC8565/MUX output control.
- Coordinated shared SPI bus access with a threading lock while continuous EEG sampling, ADC reads, DAC initialization, watchdog PWM, and TDM MUX sequencing run concurrently.

LinuxAI Raspberry Pi Voice Assistant | *Python / embedded Linux / safe command execution*

- Created a Raspberry Pi focused assistant that translates natural-language or voice-driven requests into safe Linux command plans with explanation and approval before execution.
- Integrated microphone capture through arecord, speech-to-text through whisper.cpp, command streaming, safety blocking, and Linux kernel source inspection workflows.

Artix-7 MNIST Parallel HLS Generator | *Keras-to-HLS / fixed-point acceleration*

- Built a generator that trains a compact MNIST classifier, quantizes weights, and emits a Vitis HLS-ready Artix-7 project using ap_fixed<12,4> arithmetic and four parallel MAC lanes.

Education

PhD in Computational Behavioural Neuroscience | University of Lethbridge | In progress

Current role: Research Assistant. Relevant focus: computational neuroscience, behavioural analysis, neural data systems, AI/ML hardware, embedded systems, FPGA design, Linux systems, and signal acquisition.